

Maheer Ihtejaz Moin

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EDUCATION

The University of Sydney, Sydney, NSW

Feb 2024 - Dec 2028 (Expected)

Bachelor of Advanced Computing (Honours) | Majors: Computational Data Science; Finance

Dalyell Scholar (University of Sydney high achievers program); Vice Chancellor's Scholarship

COMPETITIONS AND AWARDS

IMC Trading Algothon, Algorithmic Trading Competition

2026

Team Lead | 9th of 40 teams, top 10 finish | Python, NumPy, Pandas

- Led signal research, out-of-sample validation and release of the final trading system to a top 10 finish, scored on a hidden second year of market data no entrant saw before submission.
- Delivered \$568,609 of backtest PnL at 14.85 Sharpe and \$4,898 maximum drawdown across nine instruments over 365 trading days, by reverse-engineering each price process from its stated mechanism and pricing it in a dependency-free Python system that runs on stock CPython with no third-party packages.
- Raised mean PnL on the game-theoretic instrument to \$588,237 across 300 held-out opponent simulations, 4.4x the pooled baseline at a third of the loss rate, by fading the room's prior-day lean and discounting the estimate two standard errors so the system abstains when the edge is unmeasurable.
- Held zero risk-limit breaches, invalid positions or type errors across 365 days, 1,056 Monte Carlo scenarios and adversarial price paths, by replacing a static 3% margin with exact greedy allocation against the \$600,000 notional limit and an enforcement pass that rebuilds the book the exchange's way.
- Retired three signals fitted to noise and two strategy layers worth \$152,800 a year, by split-half testing every information coefficient, running a 200-shuffle permutation null against a Bonferroni threshold, and re-testing candidates on pre-registered seed streams.

Susquehanna International Group (SIG) Algothon, Algorithmic Trading Competition

2026

Team Lead | Top 15% of all entrants, Australia and New Zealand | Python, NumPy

- Led a four-stage research cycle from signal generation to frozen submission, finishing in the top 15% of entrants drawn from every university in Australia and New Zealand.
- Produced \$1,050 mean daily PnL at 6.43 annualised Sharpe across 51 instruments and \$114.9M of traded volume, by building a market-neutral book in pure NumPy from a ridge vector autoregression ensemble, cointegration pairs and reversal signals.
- Generated +\$262,508 cumulative PnL over a 250-day held-out test at 29 ms median decision latency, by validating the frozen model on post-freeze synthetic markets and tracking PnL attribution in a real-time monitoring terminal I built.
- Cut 100+ candidate signals to the three that shipped, by benchmarking LSTMs, Kalman filters and multi-asset cointegration against pre-registered train and holdout splits and deriving blend weights statistically.

PROFESSIONAL EXPERIENCE

2A Marketing, Sydney, NSW | Full-Stack Developer

Mar 2023 - Dec 2023

- Took the company's products and sales online for the first time, by building and deploying a React storefront.
- Let non-technical staff publish content changes without a developer, by building a Supabase and PostgreSQL backend with authentication, role-based access and a live content management layer.
- Replaced email-thread chasing of shipping paperwork, by building a portal that versions each document and pushes real-time status updates to staff and customers.
- Owned the stack end to end as the sole developer, from relational schema design and row-level access rules through to build, deployment and post-launch defect fixes.

PROJECTS

Efficient Email Filtering System, Bloom Filter | C++

- Detected duplicate emails with zero false negatives and a 0.8% false-positive rate at a 1,485-item design load, by implementing a 7-hash Bloom filter in C++ over a 15,000-bit array with custom polynomial rolling hashes written from scratch.
- Made the filter usable outside a test harness, by adding Base64 serialisation for state persistence, CLI argument parsing, and both file and live ingestion paths for real-time ADD, CHECK and SHOW operations.

Intraday Volatility Forecasting, Optiver Industry Project | R, rugarch, glmnet, data.table

- Cut QLIKE forecast error 38% against the HAR-X baseline I built, by specifying a GJR-GARCH model that won 33 of 40 stock-horizon cells across 10 US equities and four intraday horizons.
- Produced 48,600 out-of-sample forecasts with zero failed fits across 360 backtest runs, by building period-level feature caches over nine months of Optiver limit order book data, cutting repeat-fit cost enough to make the sweep tractable.
- Reduced QLIKE from 1.38 at the worst configuration to 0.181 at the best, by sweeping rolling and expanding estimation windows from 30 to 500 periods and tuning elastic-net regularisation over the HAR-X feature set.

TECHNICAL SKILLS

Languages: Java, Python, C++, C, SQL, JavaScript, R

Software Engineering: Software development lifecycle, systems architecture, REST API design, object-oriented programming, data structures and algorithms, high-performance and low-latency systems, concurrency, distributed computing, hashing and probabilistic data structures, clean code, refactoring and code review, debugging and profiling, Linux, Git/GitHub version control, Docker, Kubernetes, CI/CD (GitHub Actions), Pytest, unit and regression testing, Agile

Web, Data and Platforms: React, Node.js, Tailwind CSS, Supabase, PostgreSQL, relational schema design, REST APIs, real-time dashboards, full-stack development, Streamlit, Shiny

Statistics and Machine Learning: Time-series forecasting (ARMA-GARCH, GJR-GARCH, HAR-RV, vector autoregression), volatility modelling, ridge and elastic-net regularisation, Kalman filters, LSTM, XGBoost, Monte Carlo simulation, hypothesis testing, permutation tests, probability and statistics, NumPy, Pandas, SciPy, data.table, rugarch, glmnet

Quantitative and Trading: Trading system design, signal and alpha research, backtesting and simulation, market microstructure, limit order book (LOB) data, profit and loss (PnL) attribution, Sharpe ratio, risk management, position and notional limits, statistical arbitrage, mean reversion, cointegration, pairs trading, market-neutral portfolio construction, options pricing, portfolio optimisation, research infrastructure and tooling